# Ojasva Mishra

Z Fellow | Software Engineering | AI/ML | Quantitative Research | Robotics
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## Education

**University of Florida** — Expected May 2029
*B.S. Computer Engineering and Finance | GPA: 4.0/4.0 | YC Startup School 2026 | SAT 1590, National Merit* — *Gainesville, FL*

## Experience

**Carnegie Mellon University, Xu Lab** — Jul 2025 – Present
*Research Intern* — *Pittsburgh, PA*
- Tune PID gains so robot joints stay stable at motor torque limits, and filter attacked sensor feedback via conformal prediction.
- Validated the controllers on a Franka Emika Panda arm with Prof. Min Xu and Dr. Xiaolong Wu; **first author on 3 papers**.

**Drift (Antler-backed)** — Nov 2025 – Aug 2026
*AI/Software Engineer (Founding Team)* — *San Francisco, CA*
- Built the **Gaussian splat pipeline** that turns photos of real spaces into walkable 3D robot simulation worlds (ROS2, Gazebo).
- Built a self-healing loop that triages user-reported errors and auto-patches code: **60% more fixed** across **3,000+ errors**.
- Built core infra, internal tools, and vision-graded LLM agent evals for a robotics coding agent **1,000+ people** use.
- Traced a production Gazebo freeze to gz-transport UDP multicast failing on Ubuntu 24.04, then shipped the fix.

**Twinly** — Dec 2025 – Present
*Co-Founder and CEO (Z Fellows cohort)* — *San Francisco, CA*
- Build AI digital clones that model a user's writing, speech, and computer behavior, then run that user's tasks in parallel.
- Built the agent that lets each clone use a real computer by reading apps through OS accessibility APIs, browser DOM, and screen vision; shipped on macOS, Windows, and iOS to **1,000+ active users, including paying customers**.

**AlgoGators Investment Fund** — Sep 2026 – Present
*Quantitative Research Analyst* — *Gainesville, FL*
- Research and backtest systematic trading signals for a ~$500K AUM student fund as 1 of 2 first-years on the quant team.
- Ran a hurricane-hedged crack-spread strategy (crude vs. refined fuels): **64% win rate over 1,200 live trades**.

**EV Bots** — Apr 2025 – Feb 2026
*AI Robotics Engineer Intern* — *Chester Springs, PA*
- Built the first autonomous EV-charging robot: wrote its embedded control code (Arduino) and wired the electronics and sensors.
- Closed the perception-to-actuation loop so it finds a car and plugs in; designed a drone dock (**provisional patents**).

**PteroLabs** — Jul 2024 – Jul 2025
*Software Engineering Intern* — *Remote*
- Built a UE5 mission drone simulator with PX4, ArduPilot, and Betaflight autopilots in the loop, adopted at **UPenn**.

**iPipeline** — Jun 2025 – Aug 2025
*Software Development Intern* — *Exton, PA*
- Shipped automated test frameworks, AI agents, and developer tooling for iGO, the life insurance e-application (Python, Docker).

## Research and Publications

- Constraint-Aware Discrete-Time PID Gain Optimization for Robotic Joint Control Under Actuator Saturation. *T-RO* (accepted)
- Bounding Detector-Accepted Feedback Authority in Torque-Limited Robot Control. *ICRA 2027* (accepted)
- SCIG++: Risk-Limiting Conformal Safety Filtering for Attack-Resilient Robotic Joint Control. *IROS 2026* (under review)
- **Dream Team Engineering, NeuroSpine:** with UF Neurosurgery, build nnU-Net spine CT/MRI segmentation for bone metrics.

## Projects and Competitions

**Déjà View** | SwampHacks XII, **1st** in AI/ML and DeepSpace tracks | Live | GitHub | Video | Devpost
- Trained a 1M-Gaussian splat from COLMAP poses in 11 min and exported the scene to Meta Quest, web, and Roblox.

**Courtside** | Gator Quant Hacks 2026, Systematic Trading track | Paper | GitHub | Video
- Tested a pre-registered latency-arbitrage hypothesis on 13,084 Polymarket tennis markets: wallets trading within 3 s of a point were **profitable after fees in all 11 out-of-sample months** (Markov fair value, deflated Sharpe, block bootstrap).
- Built a causal computer-vision ball tracker that calls points a median **162 ms before the bounce** at 119.9 fps, none wrong.

**Lucid Dreams** | Gator Hack V, **1st place** | Live | GitHub
- Turned a phone video of a person into a life-size 3D replica in Meta Quest; ~290K Gaussians a frame on **12 NVIDIA B200s**.

**More projects and awards**
- **Volition** (exoskeleton concept): EEG-controlled; 1 kHz torque-limited control sim walks 8 steps from decoded intent.
- **VEX Robotics, 3x Worlds** (captain, lead programmer); **TSA, 2x National Finalist**: SAFMS, drone, VR, controls portfolios.

## Technical Skills

**Languages:** Python, C++, C, Rust, Java, C#, Swift, TypeScript, JavaScript, SQL, MATLAB
**AI and infrastructure:** PyTorch, OpenCV, CoreML, LLM agents, Claude Code, Codex, Cursor, Linux, Docker, AWS, GCP, CI/CD
**Robotics:** ROS2, Gazebo, MuJoCo, Isaac Sim, PX4, ArduPilot, SLAM, PID control, Gaussian splatting (gsplat, COLMAP)
**Quant:** backtesting, systematic signals, Markov models, deflated Sharpe, block bootstrap, NumPy, pandas, SciPy
**Hardware:** Arduino, microcontrollers, RTOS, PCB and circuit design, motor drivers, I2C/SPI/UART, oscilloscopes, soldering, CAD